

ECON 100B: Macroeconomics

Class 2: Measuring the Macroeconomy

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January 22, 2026

Agenda

- Announcements
- Key concepts and Figure 1.1 for the growth part (Chapter 1)
- Measuring the Macroeconomy (Chapter 2)

Announcements

- Our **FINAL EXAM** is tentatively scheduled for Monday, May 11, 3–6 PM. We will use 90 minutes
- CalCentral may state nothing, or it may state something else that is incorrect
- Our **MIDTERM EXAM** will be here in-class on Thursday, March 5. We will use 60 minutes
- DSP students: Please schedule proctoring or plan to use our separate 150% time classroom if that works & you want to. Details forthcoming

Course videos

- Please navigate to bCourses: Media Gallery
- These are the Course Capture videos
- I think these work, and I'd like to leave it at that, but let me know

ECON 000 on Ed Discussion

- Kind of like a subreddit for ECON majors and departmental advisors and other staff
- Open to anybody with a CalNet ID

<https://edstem.org/us/join/JSg8Dw>

The screenshot shows the Ed Discussion forum interface. At the top, the browser address bar displays `edstem.org/us/courses/40590/discussion/`. Below the browser, there's a purple header bar with the 'ed' logo and the text 'Econ 000 – Ed Discussion'. On the left side, there's a sidebar with a 'New Thread' button and a list of courses: COMPSS 202 LEC 001, ECON 140 LEC 001, ECON 157 LEC 001, and 'Econ 000' which is highlighted with 989 members. Below the courses, there's a list of categories: General, Declared Majors, Prospective Majors, Econ Classes, Opportunities/Events, Weekly Announcements, Social, Alumni, and Commencement. The main content area on the right shows a search bar and a list of discussion threads. The threads include 'Advising Services for Week 2: 9/2 - 9/6', 'Seeking ECON 136 and ECON 101A UGSIs, Fall 2024', 'Fall 2024 Econ 197 / CPT Application', 'Berkeley Economic Review (BER) Application', 'Undergrad Econ Association (UEA) Infosession Tonight @ 8PM', 'Undergrad Econ Association Infosession Tonight!', 'IndEng 120', and 'Declaring major'. Each thread shows its category, author, time since posted, and engagement metrics like replies and likes.

ed Econ 000 – Ed Discussion

New Thread

COURSES

COMPSS 202 LEC 001

ECON 140 LEC 001

ECON 157 LEC 001

Econ 000 989

[INACTIVE] Introduction ...

CATEGORIES

- General
- Declared Majors
- Prospective Majors
- Econ Classes
- Opportunities/Events
- Weekly Announcements
- Social
- Alumni
- Commencement

Search

Filter

- Advising Services for Week 2: 9/2 - 9/6
General Economics Staff STAFF 3d 1
- Seeking ECON 136 and ECON 101A UGSIs, Fall 2024
Declared Majors Isabel Burle STAFF 1mth
- Fall 2024 Econ 197 / CPT Application
Declared Majors Economics Staff STAFF 1mth 2 1

Show 8 more

This Week

- Berkeley Economic Review (BER) Application
Opportunities/Events Inseo Yang 1h 3
- Undergrad Econ Association (UEA) Infosession Tonight @ 8PM
General Mildred Flores STAFF 3h 2
- Undergrad Econ Association Infosession Tonight!
Opportunities/Events Anonymous 3h 1
- IndEng 120
General Anonymous 4h 1
- Declaring major
General Anonymous 4h 1

Outline of Macroeconomics

- Chapter 2 shows how we measure income, $Y = C + I + G + NX$
- Chapters 3–6 discuss long-run growth
- Chapters 7–8 talk about inflation and unemployment
- Chapters 9–14 discuss short-run fluctuations, fiscal policy, monetary policy, the Phillips Curve, the Fed
- Chapters 16 & 17 focus on Consumption and Investment
- Chapter 18 looks at Government
- Chapters 19 & 20 address international trade and finance, exchange rates

Learning objectives in Chapter 2

- Income is a good measure of human well-being
- Average income = income per person
- Gross domestic product (GDP) is our most ubiquitous measure of income in a country or geographic region. It's easier to measure especially in developing regions
- We want to measure **real GDP per person**, usually $y = \frac{Y}{L}$
- GDP is usually measured by expenditures, $Y = C + I + G + NX$
- Real GDP is nominal GDP divided by a price index: $Y^R = \frac{Y^N}{P}$
- Measuring the price index is tricky but doable across time & space

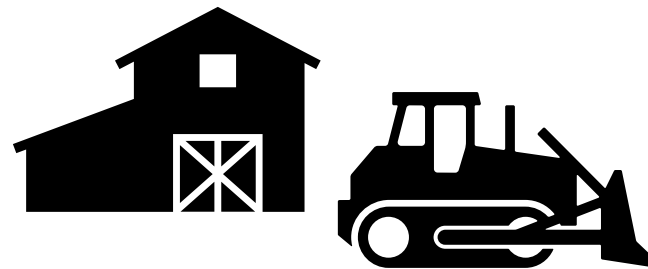
Lesson 1

- In the long run, economic well being is determined by factor inputs:

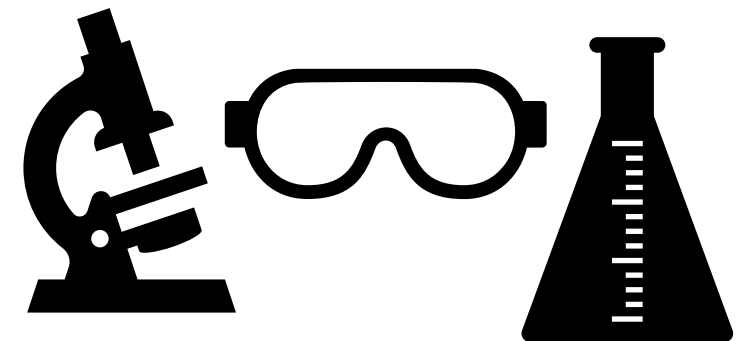
Labor



Capital

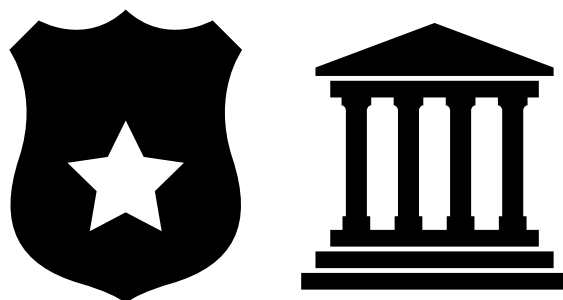


Technology

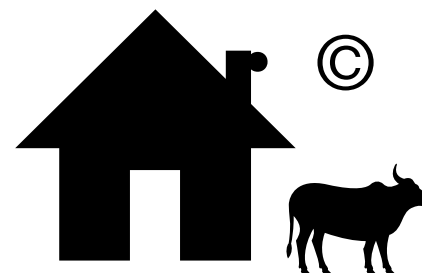


- For economic agents to produce these, typically we need “institutions” or “social infrastructure”

Rule of law



Property rights



Protection from government

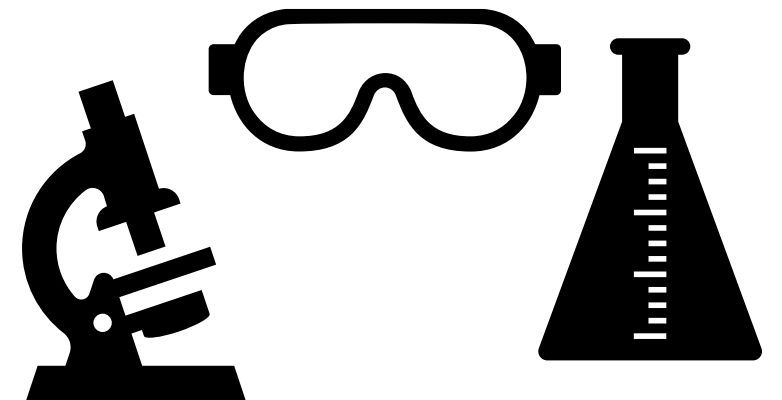


Lesson 2

- Improvements in average labor productivity are the key source of **long-run growth**,



- and capital accumulation and technological progress are the key sources of improvements in average labor productivity.

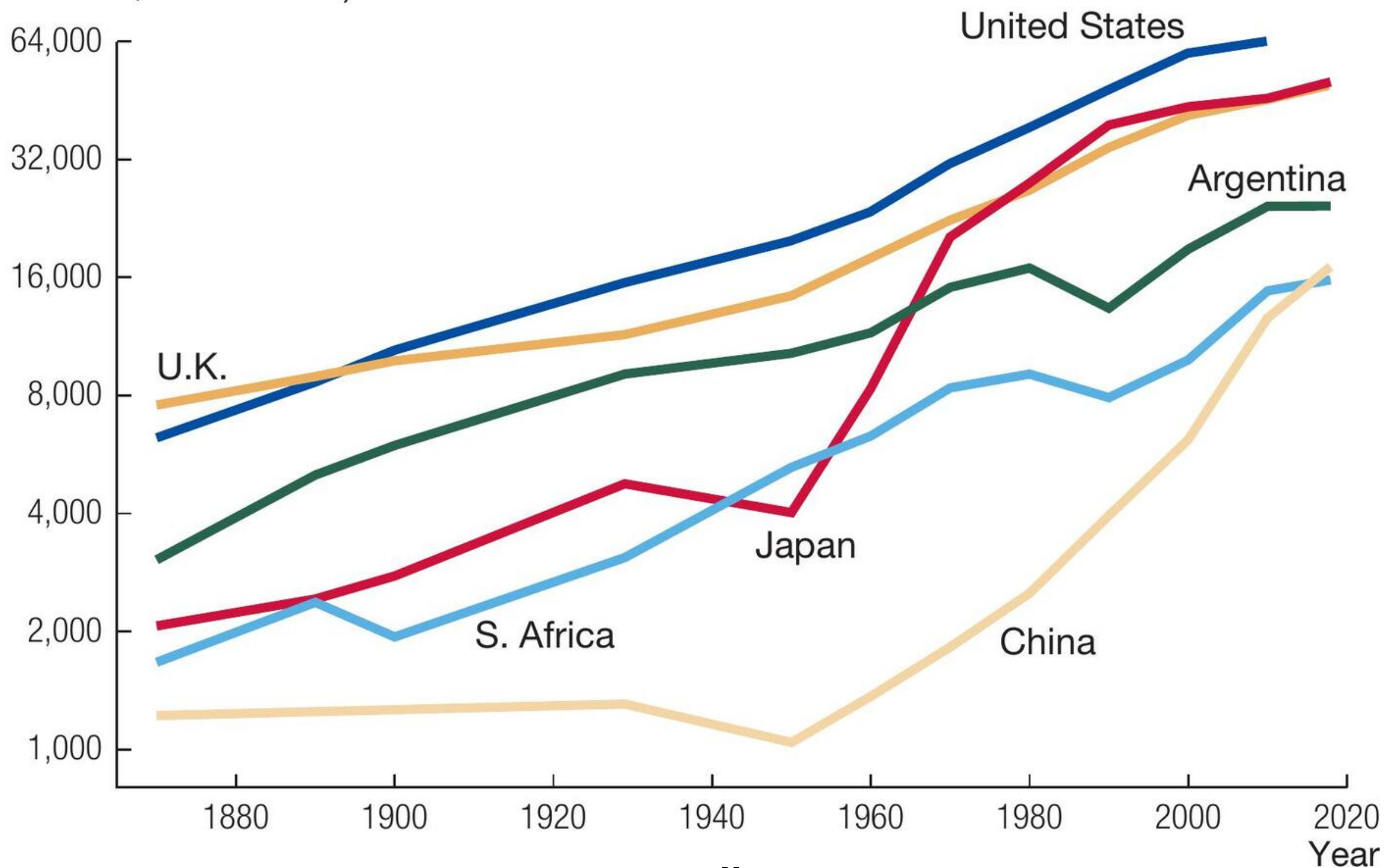


Growth in what? Fluctuations in what?

- Human well-being
- But how do we measure that? Maybe just ask people?
 - How much do you earn or receive in profits, interest, or dividends?
 - How much do you have in assets and investments?
 - How happy are you? Maybe it's a function of these things above?
- After the Great Depression, and long before the monthly Current Population Survey (unemployment), we had [NBER](#), [Simon Kuznets](#), and **National Income Accounting** that could tell us total income and thus **average income = income per person**

**Figure 1.1 Why are some countries rich while some are poor?
Why do some grow faster than others? Why is there any growth?**

GDP per person
(ratio scale, 2022 dollars)



National income accounting produces estimates of GDP

- Gross Domestic Product (GDP) measures the total value of production in an economy in a year
- Everything produced must be purchased (or be placed in an inventory), that is, bought and sold
- When you buy something, someone gets paid — and the proceeds get divvied up between everyone doing the work: workers, managers, landlords, etc.
- So total production equals total expenditures equals total income

An example

Homer and Marge own a tangerine farm with workers

- Production: In 2005, their workers picked 100 tangerines whose market value was \$1 per tangerine
- Expenditure: In 2005, their fruit stand sold 100 tangerines to consumers at \$1 per tangerine
- Income: In 2005, Homer and Marge netted a profit of \$33 while their workers earned \$67

Measured each way, GDP for the farm is \$100 in 2005

Note there's no saving and investment. These are tangerines, so they must be consumed

The most commonly referenced measure of GDP is expenditures grouped by function

The national income identity

Income equals	$Y =$
consumption	C
plus investment	$+ I$
plus government purchases	$+ G$
plus net exports	$+ NX$

Count up value of the new cars and soybeans (C) plus
new construction equipment (I) plus
new roadways (G) plus
soybeans exports minus imported cars (NX)

Note what happens when there aren't bells and whistles like government and foreign trade

The national income identity

Income equals $Y =$
consumption C
plus investment $+ I$

Now consider that investment goods represent **saving**

Things then look more familiar: Income = Spending + Saving
 $Y = C + I$

	Total (billions of dollars)	Share of GDP (percent)	Per person (dollars)
Gross domestic product	14,440	100.0	47,010
Consumption	10,130	70.2	32,970
Motor vehicles and parts	340	2.4	1,110
Food	780	5.4	2,550
Housing	1,840	12.7	6,000
Medical care	1,550	10.7	5,060
Investment	2,140	14.8	6,950
Structures (nonresidential)	610	4.2	1,980
Equipment and software	1,080	7.5	3,530
Residential	480	3.3	1,550
Government purchases	2,880	19.9	9,380
National defense	740	5.1	2,400
Net exports	−710	−4.9	−2,300
Exports	1,830	12.7	5,960
Imports	2,540	17.6	8,260

TABLE 2.1 The Expenditure Approach to U.S. GDP in 2008

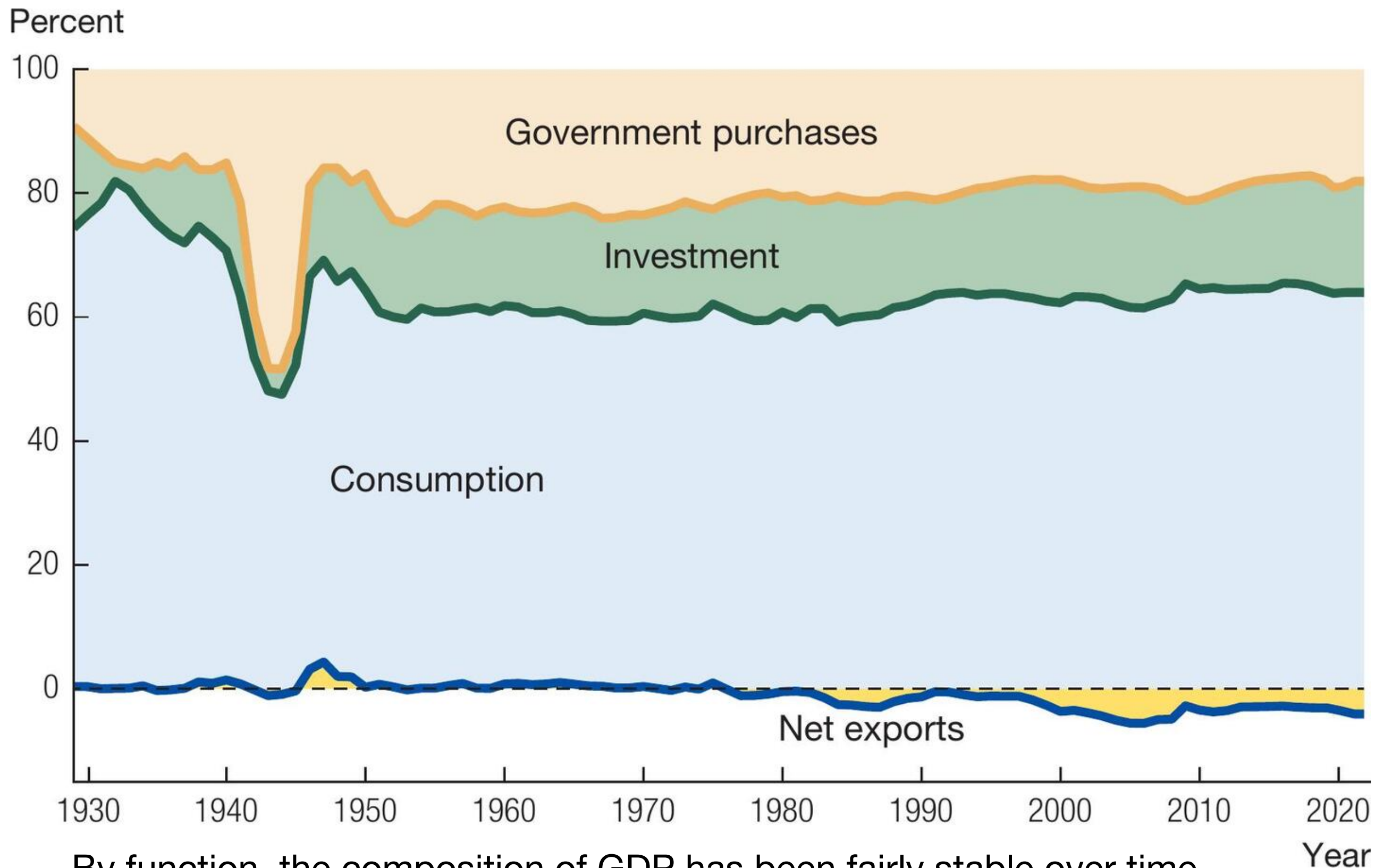
Macroeconomics, 2nd Ed
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TABLE 2.1

The Expenditure Approach to U.S. GDP in 2022

	Total (trillions of dollars)	Share of GDP (percent)	Per person (dollars)
Gross domestic product	25.5	100.0	76,300
Personal consumption expenditures	17.4	68.2	52,000
Motor vehicles and parts	0.7	2.7	2,200
Food	1.3	5.1	3,800
Housing	3.0	11.8	9,000
Medical care	2.7	10.6	8,200
Gross private domestic investment	4.6	18.0	13,900
Structures (nonresidential)	0.6	2.4	1,900
Equipment	1.3	5.1	4,000
Intellectual property products	1.4	5.5	4,100
Residential	1.1	4.3	3,400
Government purchases	4.4	17.3	13,300
National defense	0.9	3.5	2,800
Net exports of goods and services	−1.0	−3.9	−2,900
Exports	3.0	11.8	8,900
Imports	4.0	15.7	11,800

Source: U.S. Department of Commerce, Bureau of Economic Analysis, Table 1.5.5, bea.gov.



By function, the composition of GDP has been fairly stable over time

But World War II was a time of vastly expanded government purchases

Income shares of GDP have remained fairly stable over time

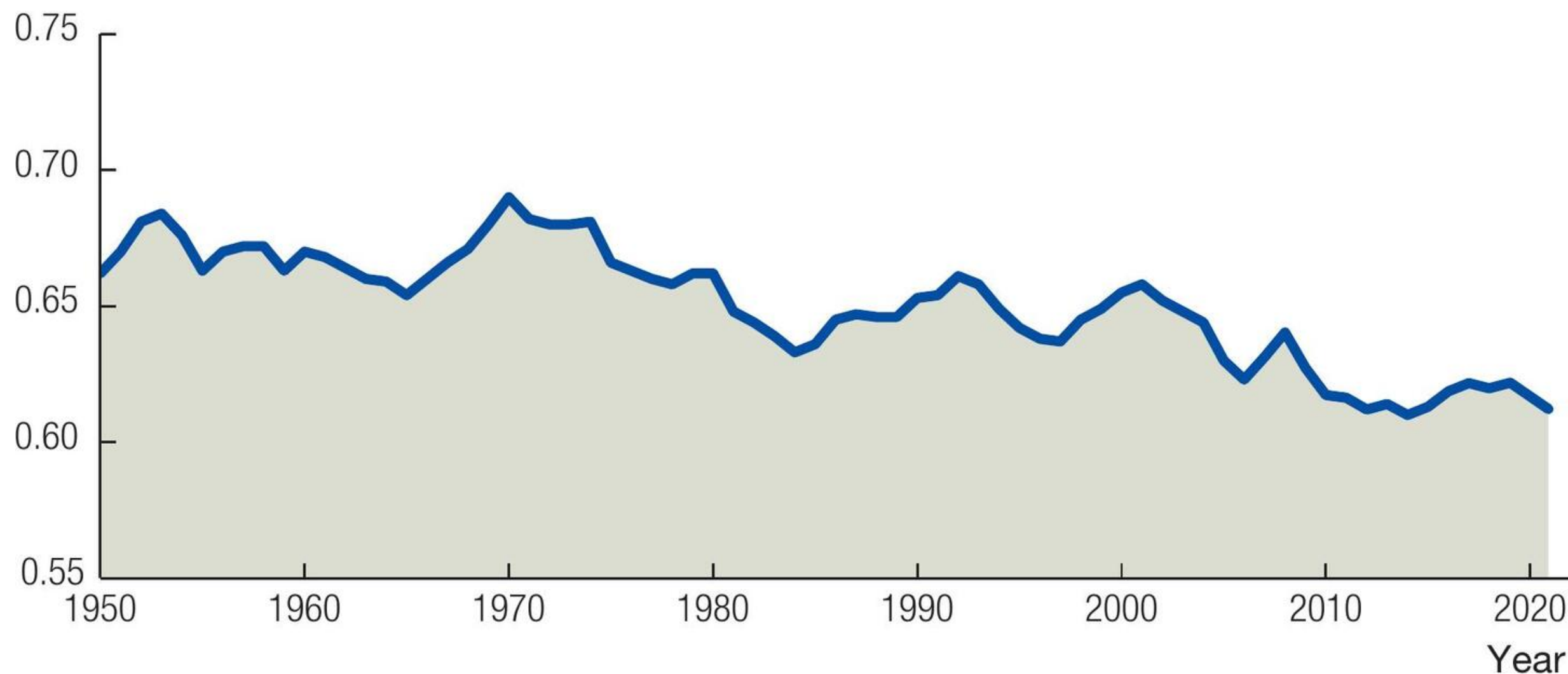
Broadly, there are two factors of production:

Labor: workers earning wages, including business owners

Capital: machinery, buildings, equipment

Labor has earned about two-thirds of GDP

Labor Share of GDP



GDP is “value added”

- The macroeconomy is filled with many interdependent consumers and firms, not just one Homer and Marge, Inc.
- When measuring the total value of production, we can't **double count** by measuring intermediate production. We count only final goods *OR* value added
- Example:
 - A steel company produces \$10 million worth of steel
 - A car company buys all of it and produces \$100 million in (steel) cars
 - Total GDP is \$100 million: either the \$100 million in final goods (the cars) or the \$10 million in steel production value added plus the value added by the car company, which is $100 - 10 = \$90$ million

Limitations to the GDP measure

- GDP only measures market transactions
 - Anything you “pay yourself to do,” like housekeeping, own child-care, etc. does not get measured
 - This is a huge problem in countries with large black markets
 - Sales of used goods are not counted because they are not produced, only transferred between individuals
- GDP does not measure changes in health or in the environment
- Separating prices and quantities out of a measure of total value is difficult

Measuring REAL well-being means you have to measure prices correctly

- Why? We can always count currency-units (dollars) of income,
 - but we want to know how many units of goods and services that income purchases,
 - and then we want to compare those quantities over space & time
- To help us find quantities, a familiar accounting principle will hold:
 - Production in dollars = Price x Quantity
 - Nominal GDP = Price Level x Real GDP $Y^N = P \times Y^R$
- So the key is how the Price Level changes over space & time

Measuring prices is harder than you might think

- The problem is that prices of different goods can change in different ways over time and space ...
- So it's difficult to measure THE price level; there are several candidate choices rather than just one
- For example, suppose the price of computers fell by 2% but the price of food rose by 5% — What is the new price level?
- That's a tough question to answer
- As the textbook explains in Chapter 2, a good choice for the price level is a geometric average between prices measured in the past and measured today

The bottom line: we measure **REAL GDP**

- Using the techniques described in the textbook, economists have produced measures of real GDP that are comparable over time & space
- For example, our dataset shows real GDP per capita in the U.S. was about \$30,000 in 2000, while it was about \$10,000 in 1950
- That really means: our best guess is that the average resident in 2000 enjoyed a real standard of living 3 times as high as that of the average resident in 1950
- For comparison: nominal GDP per capita in 1950 was \$2,000
- We have removed the effects of overall inflation and changes in relative prices between goods
- What to remember? Comparisons across time **or space...** are meaningful when they are done with modern methods

Textbook's discussion of Chinese and U.S. GDP when adjusted for purchasing power parity (PPP)

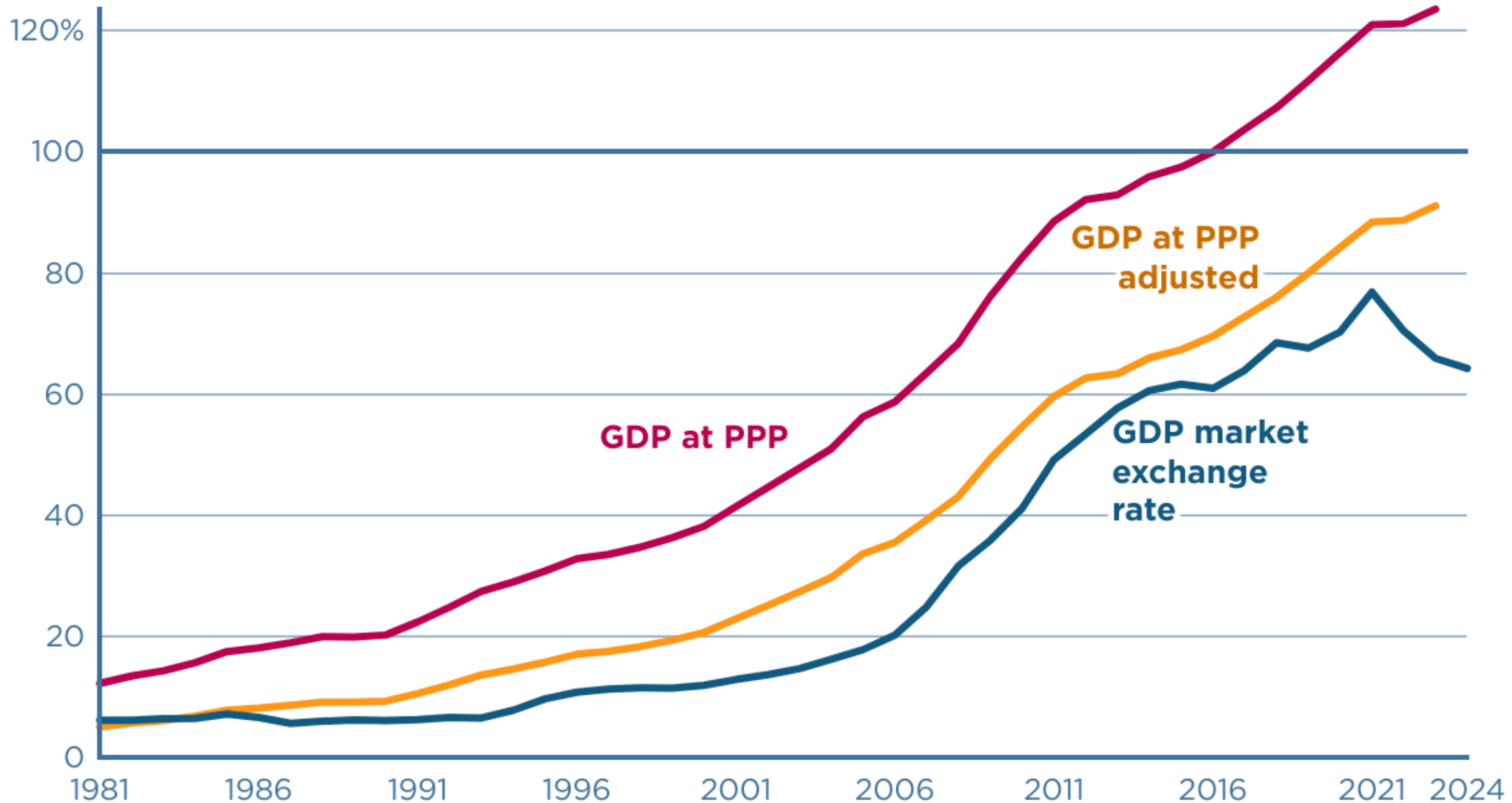
- Price levels are not the same across international borders!
- Adjusting for exchange rates alone will miss this, and it would make China's GDP look much smaller than it probably is
- Because domestic prices in China are “lower” than U.S. prices — meaning Chinese incomes buy more real goods — China's real PPP-adjusted GDP is larger than it would appear
- Rather than 33% smaller than U.S. GDP based on the exchange rate alone, Chinese GDP in 2019 was perhaps about 2% smaller
... maybe ...

Figure 1

Although China's GDP (at PPP) is ahead of the United States, its economic power may still be lower

[Dec 18, 2025 article at PIIE](#)

China's GDP-based measures of relative economic power, as a % of US GDP, 1981-2024



PPP = purchasing power parity



Source: [Penn World Tables](#) for GDP at PPP, [World Bank World Development Indicators](#) for GDP at market prices, and Federal Reserve Economic Data (FRED) for [China](#) and [US](#) populations.

On to growth: A sketch of the U.S. around 1900

- Life expectancy at birth, the average length of life starting from birth, was about 50 [today: 77]
- One out of every 10 infants died before his or her first birthday [today: 7 out of every 1,000]
- 90 percent of households did not have electricity, refrigerator, telephone, or a car [today most do]
- Fewer than 10 percent of adults had graduated from high school [today: 85 percent]

A sobering comparison

	USA 1900	USA 2000	Kenya 2000
Life expectancy at birth	50	77	50
Infant mortality	0.1	0.007	0.06
Real GDP per capita (1990 \$)	\$4,100	\$28,000	\$1,000

Healthier = wealthier, with heterogeneity

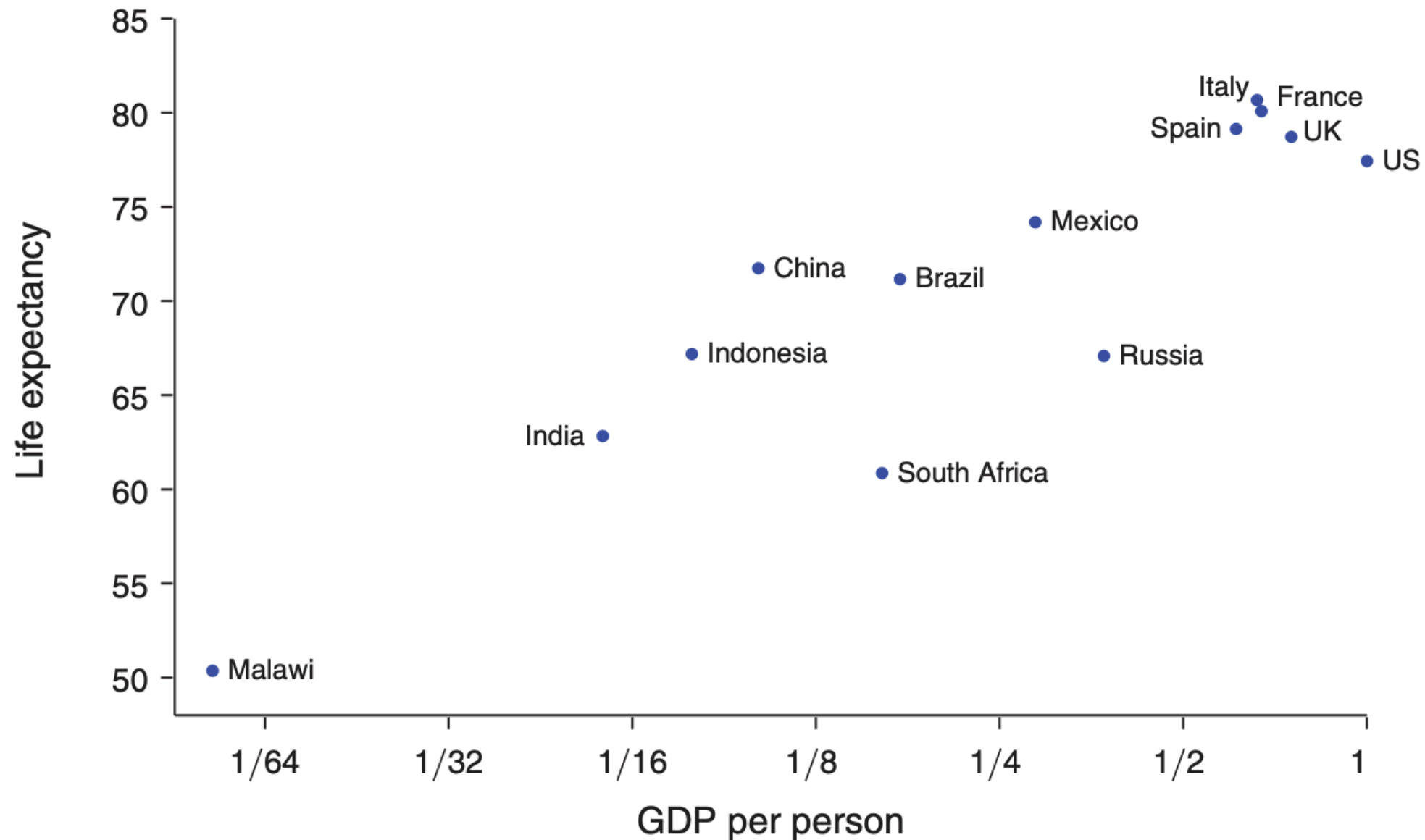


FIGURE 4. LIFE EXPECTANCY

Note: Life expectancy at birth in each country is measured as the sum over all ages of the probability of surviving to each age, using life tables from the World Health Organization.